

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
getwd()
```

```
## [1] "/home/guest/GITclone/EDE_Fall2025"
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.0      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(zoo)
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##      as.Date, as.Date.numeric
```

```
library(trend)

my_theme <-
  theme_dark(base_size = 14) +
  theme(axis.text = element_text(color = "firebrick"),
        legend.position = "top")
theme_set(my_theme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1
Garinger2010 <-
  read.csv("/home/guest/GITClone/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv")
Garinger2011 <-
  read.csv("/home/guest/GITClone/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv")
Garinger2012 <-
  read.csv("/home/guest/GITClone/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv")
Garinger2013 <-
  read.csv("/home/guest/GITClone/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv")
Garinger2014 <-
  read.csv("/home/guest/GITClone/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv")
Garinger2015 <-
  read.csv("/home/guest/GITClone/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv")
Garinger2016 <-
  read.csv("/home/guest/GITClone/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv")
Garinger2017 <-
  read.csv("/home/guest/GITClone/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv")
Garinger2018 <-
  read.csv("/home/guest/GITClone/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv")
Garinger2019 <-
  read.csv("/home/guest/GITClone/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv")

Garinger_df <-
full_join(Garinger2010, Garinger2011, by = NULL, copy = FALSE, keep = NULL)%>%
  full_join(Garinger2012)%>%
  full_join(Garinger2013)%>%
  full_join(Garinger2014)%>%
  full_join(Garinger2015)%>%
  full_join(Garinger2016)%>%
```

```
full_join(Garinger2017)%>%
full_join(Garinger2018)%>%
full_join(Garinger2019)
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to “Date”.
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
# 3
Garinger_df$Date <-
mdy(Garinger_df$Date)
# 4
Garinger_df_processed <-
  Garinger_df %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)
# 5
Days <- as.data.frame(seq.Date(from=as.Date("2010/01/01"),
                                to=as.Date("2019/12/31"),
                                by="day"))

colnames(Days) <-
  "Date"

# 6
GaringerOzone <- left_join(Days, Garinger_df_processed)
```

```
## Joining with 'by = join_by(Date)'
```

Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

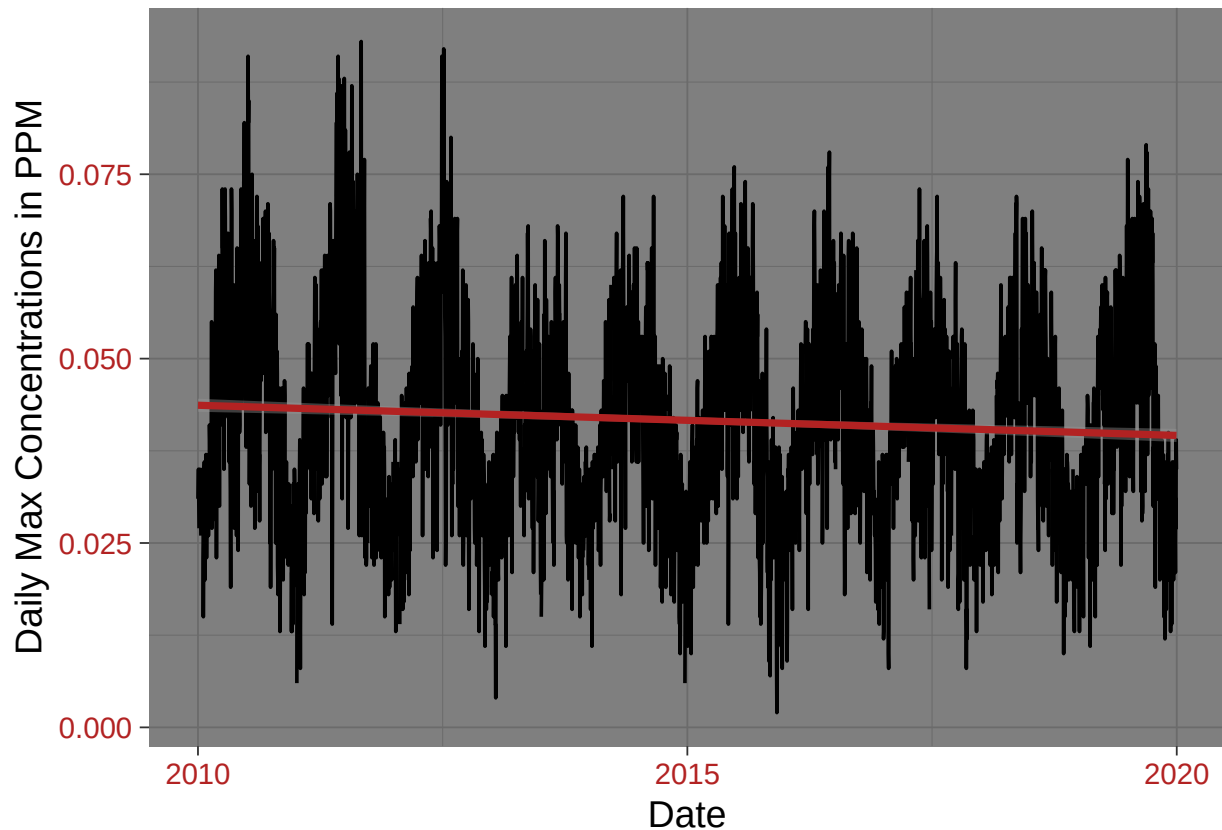
```
#7
Ozone_Conc_Plot <-
ggplot(GaringerOzone, aes(x= Date,
                           y= Daily.Max.8.hour.Ozone.Concentration)) +
  geom_line()+
  geom_smooth(method = 'lm',
              color = "firebrick")+
  
```

```
ylab("Daily Max Concentrations in PPM")+
  my_theme

Ozone_Conc_Plot
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: This plot suggests a seasonal trend in ozone concentrations over time. Daily maximum ozone concentrations are lowest in winter months and highest in summer months. Over a period of ten years, ozone concentrations appear to be slightly decreasing.

Time Series Analysis

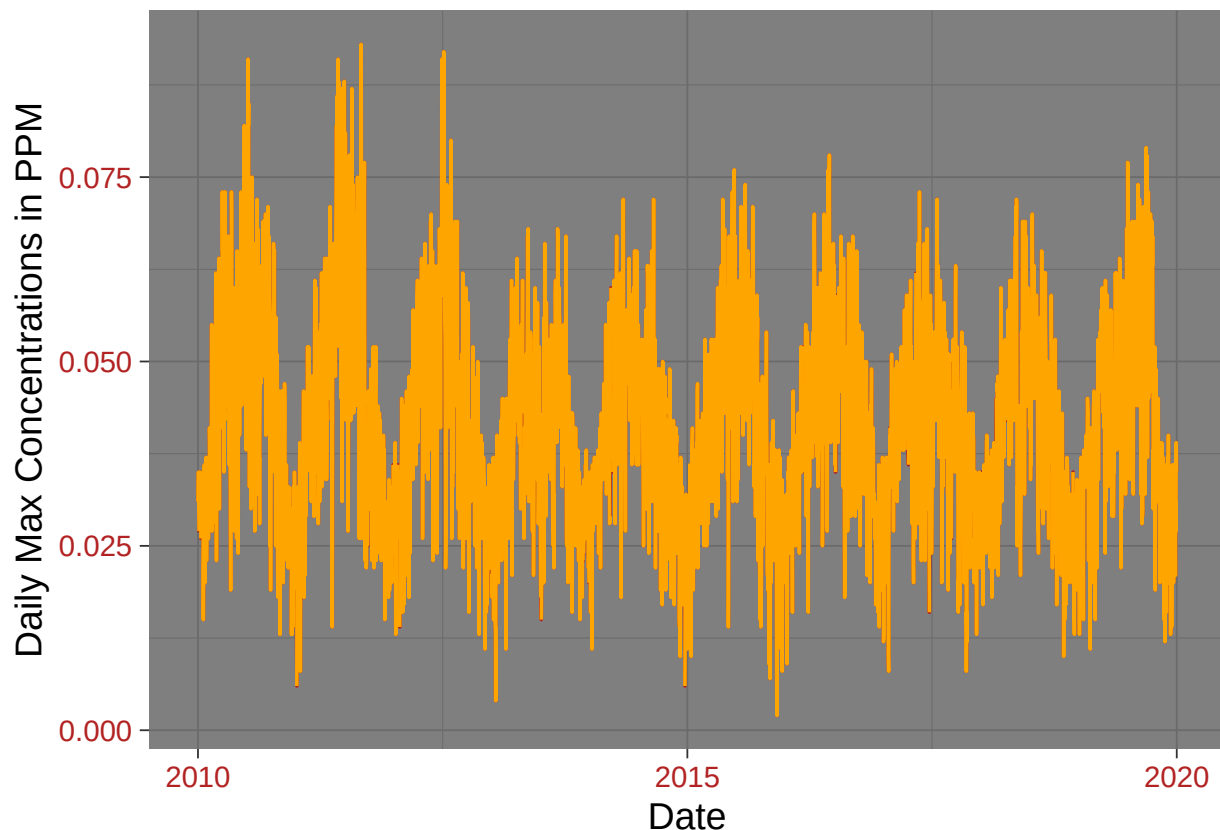
Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
GaringerOzone_clean <-
  GaringerOzone %>%
  mutate( Daily.Max.8.hour.Ozone.Concentration.clean = zoo::na.approx(Daily.Max.8.hour.Ozone.Concentration,
  summary(GaringerOzone_clean$Daily.Max.8.hour.Ozone.Concentration.clean)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00200 0.03200 0.04100 0.04151 0.05100 0.09300
```

```
#Note the NA is gone
GaringerOzone_clean.plot <-
ggplot(GaringerOzone_clean) +
  geom_line(aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration.clean), color = "firebrick") +
  geom_line(aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration), color = "orange") +
  ylab("Daily Max Concentrations in PPM") +
  my_theme
GaringerOzone_clean.plot
```



Answer: We didn't use a piecewise constant because the ozone concentrations appear to be seasonal, thus it is more likely that missing data points lie somewhere between the Ozone values of a past or future date. A spline interpolation is less effective than a linear interpolation in this dataset because splines are more effective in datasets with more variability and unpredictability.

Again, the ozone data is seasonal and thus a more constant, linear increase/decrease would be more realistic.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9
GaringerOzone.monthly <-
  GaringerOzone_clean %>%
    mutate(Year = year(Date), Month = month(Date)) %>%
    group_by(Year, Month) %>%
    summarise(meanOzone = mean(Daily.Max.8.hour.Ozone.Concentration.clean)) %>%
    mutate( Date = my(paste0(Month, "-", Year)))
```

```
## 'summarise()' has grouped output by 'Year'. You can override using the
## '.groups' argument.
```

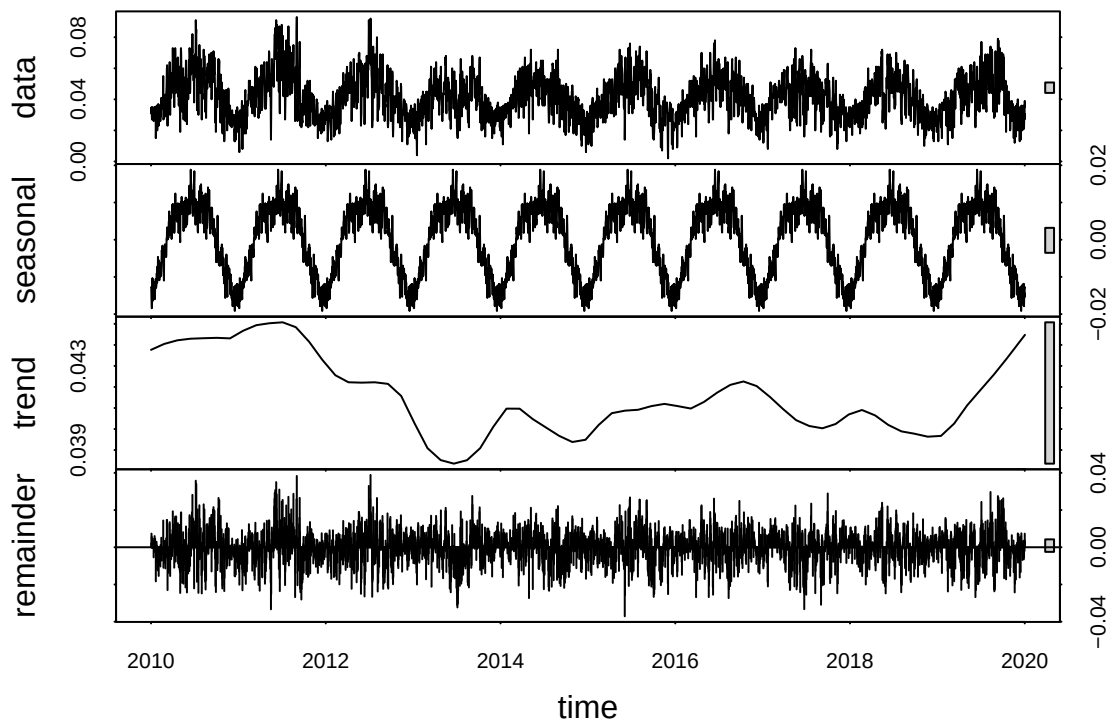
10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

```
#10
GaringerOzone.daily.ts <-
  ts(GaringerOzone_clean$Daily.Max.8.hour.Ozone.Concentration.clean,
      start = c(2010,1),
      frequency = 365)

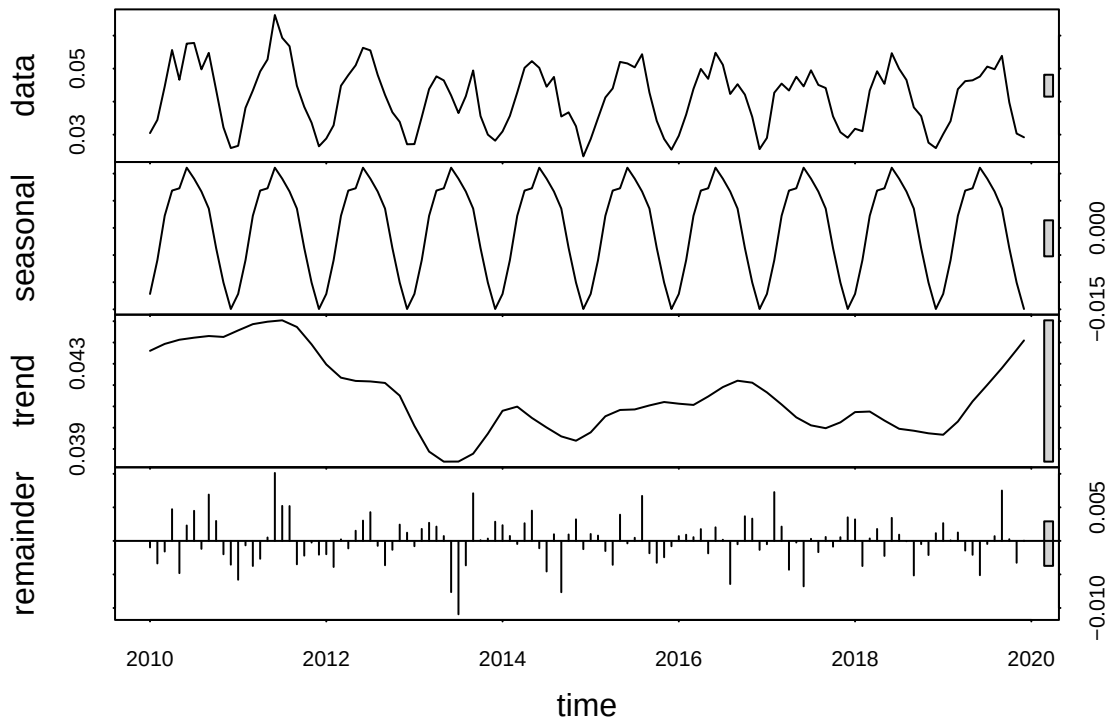
GaringerOzone.monthly.ts <-
  ts(GaringerOzone.monthly$meanOzone,
      start = c(2010,1),
      frequency = 12)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11
Garinger.daily.decomp <- stl(GaringerOzone.daily.ts,
                             s.window = "periodic")
plot(Garinger.daily.decomp)
```



```
Garinger.monthly.decomp <- stl(GaringerOzone.monthly.ts,
                               s.window = "periodic")
plot(Garinger.monthly.decomp)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

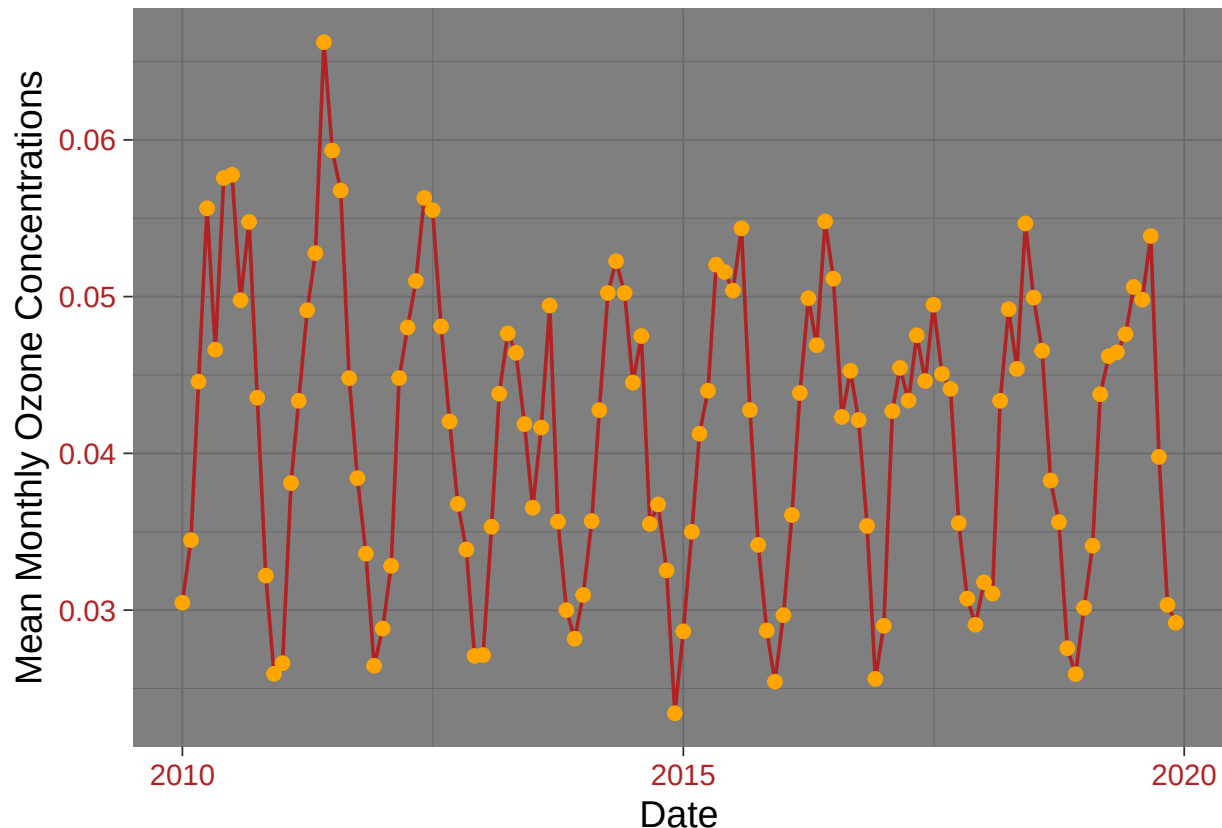
```
#12
monthly_Ozone_trend <- Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)
```

Answer: The seasonal Mann-Kendall trend analysis is most appropriate because our dataset definitely exhibits a seasonal pattern. A linear regression, regular Mann-Kendall analysis, and Spearman Rho all do not account for seasonality.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
mean.monthly.ozone.plot <-
  ggplot(GaringerOzone.monthly, aes(x = Date, y = meanOzone)) +
  geom_line(color = "firebrick") +
  geom_point(color = "orange") +
  ylab("Mean Monthly Ozone Concentrations") +
  my_theme

mean.monthly.ozone.plot
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: Mean monthly ozone concentrations exhibit a similar trend as daily ozone concentrations over the same time period. In the span of ten years, ozone concentrations are decreasing slightly. The seasonal Mann-Kendall test reflects this with a tau value of -0.143 (ozone decreasing) and a statistically significant p-value of 0.046724 (less than 0.05). In the context of the research question, our data points to the conclusion that ozone concentrations have changed over the 2010s at this research station. (Seasonal Mann-Kendall result: tau = -0.143, 2-sided pvalue = 0.046724)

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

#15

```
Garinger_monthly_Components <- as.data.frame(Garinger.monthly.decomp$time.series[,1:3])

Garinger_monthly_Components <- mutate(Garinger_monthly_Components,
  Observed = GaringerOzone.monthly$meanOzone,
  Date = GaringerOzone.monthly$Date)
```

```
summary(Garinger_monthly_Components)
```

```
##      seasonal      trend      remainder      Observed
## Min.      :-0.014935 Min.      :0.03841 Min.      :-1.094e-02 Min.      :0.02342
## 1st Qu.: -0.006976 1st Qu.:0.04013 1st Qu.: -1.987e-03 1st Qu.:0.03380
## Median : 0.002895 Median :0.04108 Median : 8.770e-05 Median :0.04335
## Mean    : 0.000000 Mean    :0.04150 Mean    :-1.152e-05 Mean    :0.04149
## 3rd Qu.: 0.006982 3rd Qu.:0.04247 3rd Qu.: 2.049e-03 3rd Qu.:0.04915
## Max.    : 0.011093 Max.    :0.04504 Max.    : 1.013e-02 Max.    :0.06623
##
##      Date
## Min.      :2010-01-01
## 1st Qu.:2012-06-23
## Median :2014-12-16
## Mean      :2014-12-16
## 3rd Qu.:2017-06-08
## Max.      :2019-12-01
```

```
Garinger_comp_ts <- ts(Garinger_monthly_Components$Observed,
  start=c("2010", "1"),
  frequency=12)
```

```
Garinger_comp_ts
```

```
##      Jan      Feb      Mar      Apr      May      Jun
## 2010 0.03046774 0.03446429 0.04458065 0.05563333 0.04661290 0.05756667
## 2011 0.02661290 0.03810714 0.04335484 0.04913333 0.05277419 0.06623333
## 2012 0.02882258 0.03282759 0.04480645 0.04803333 0.05100000 0.05630000
## 2013 0.02712903 0.03532143 0.04380645 0.04765000 0.04641935 0.04186667
## 2014 0.03096774 0.03567857 0.04275806 0.05023333 0.05225806 0.05023333
## 2015 0.02864516 0.03500000 0.04125806 0.04400000 0.05203226 0.05156667
## 2016 0.02967742 0.03606897 0.04385484 0.04990000 0.04690323 0.05480000
## 2017 0.02900000 0.04269643 0.04545161 0.04336667 0.04753226 0.04461667
## 2018 0.03177419 0.03105357 0.04335484 0.04920000 0.04538710 0.05466667
## 2019 0.03014516 0.03410714 0.04377419 0.04620000 0.04645161 0.04760000
##      Jul      Aug      Sep      Oct      Nov      Dec
## 2010 0.05777419 0.04977419 0.05476667 0.04354839 0.03220000 0.02593548
## 2011 0.05932258 0.05677419 0.04480000 0.03841935 0.03360000 0.02645161
## 2012 0.05551613 0.04809677 0.04203333 0.03677419 0.03386667 0.02708065
## 2013 0.03653226 0.04164516 0.04943333 0.03564516 0.03000000 0.02817742
## 2014 0.04451613 0.04748387 0.03550000 0.03674194 0.03253333 0.02341935
## 2015 0.05038710 0.05435484 0.04276667 0.03416129 0.02870000 0.02543548
## 2016 0.05114516 0.04232258 0.04526667 0.04212903 0.03536667 0.02561290
## 2017 0.04948387 0.04506452 0.04411667 0.03554839 0.03073333 0.02906452
## 2018 0.04993548 0.04654839 0.03826667 0.03561290 0.02756667 0.02591935
## 2019 0.05061290 0.04980645 0.05386667 0.03977419 0.03033333 0.02919355
```

```
#16
```

```
Garinger_trend <- Kendall::MannKendall(Garinger_comp_ts)
```

```
monthly_Ozone_trend
```

```
## tau = -0.143, 2-sided pvalue =0.046724
```

```
Garinger_trend
```

```
## tau = -0.0594, 2-sided pvalue =0.33732
```

Answer: The Seasonal Mann-Kendall test has a tau of -0.143, meaning that ozone levels are slightly decreasing over time. The result is also statistically significant with a 2-sided p-value of 0.046724 (less than 0.05), meaning that we reject the null hypothesis that there is no trend in ozone levels over time. The non-seasonal Mann-Kendall finds that this data has a tau value of -0.0594, meaning again that the ozone levels are slightly decreasing over time. However, the Mann-Kendall test returned a 2-sided p-value of 0.33732, meaning that the results are not statistically significant. Thus, this second test indicates that we accept the null hypothesis that there is no trend in ozone levels over time.